Bit-wise XOR

0 xor 0 = 0
0 xor 1 = 1
1 xor 0 = 1
1 xor 1 = 0

XOR **b1 + b2 mod 2**

0 + 0 mod 2 = 0
0 + 1 mod 2 = 1
1 + 0 mod 2 = 1
1 + 1 mod 2 = 0

A	B	A XOR B	A XOR B XOR B
0	0	0	0
0	1	1	0
1	0	1	1
1	1	0	1

A xor B xor B = A

Ci = Mi xor Ki (the ith bit)

Ci xor Ki = Mi xor Ki xor Ki -> Mi

Block
DES 64 bits

AES 128-bit block

If the plantext < 128 bits → Padding
Big message > 128 bits: divide message into blocks of 128 bits (AES)
Apply AES encryption on each block

DES was designed with 64 bit key

NSA No Such Agency , National Security Agency

ECB electronic code book

Block cipher

M = M1, M2, …, Mn

M1 C1= E(K, M1),

M2 C2 = E(K, M2),
...,
Mn Cn = E(K, Mn)

C2 C3 C5 C4 Cn… C1

C1 C2 C3 C4
C1 C4 C3 C2

Encrypt the message 1000 times, get same ciphertext

Secure Encryption??????

The key can not be brute forced.
K size is 1024 bits

Attacker cannot recover the message

Attack at noon.
3$5x9022noon.

Ciphertext should leak no information regarding the message!!!

Let's examine the ECB (Electronic Code Book)

Encrypt each block in separately.

1. Ciphertext blocks tied to each other → cannot change order
2. Same message → different ciphertext everytime

Error propagation

Nonce: Number use once

CBC Cipher block chaining

Can be used as stream cipher

Cipher Feedback Mode

Precompute the key bit-stream

Output Feedback Mode

Bit-stream can be computed in parallel

Counter Mode

Same massage in → different ciphertexts

Error Propagation